

Introduction to Multivariate Statistics

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Introduction

- Applied multivariate statistics overlaps and shares concepts with regression, data visualization, data mining, and machine learning.
- **Multivariate statistics** is the study of the outcomes of multiple variables simultaneously.
 - Sometimes they are studied with assumed dependent variables (supervised learning).
 - Other times, techniques are used to uncover patterns or reduce dimensionality of observations without assuming any dependent variables (unsupervised learning).

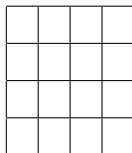
Key Areas

- Dimension reduction.
- Clustering.
- Classification.
- Note: *These methods may be combined with other methods to improve overall understanding or prediction accuracy.*

Basics of Linear Algebra

Recall: Matrix in $\mathbb{R}$

- In $\mathbb{R}$, a **matrix** is simply a vector that has additional attributed called dimensions.
- Instead of only having one-dimensional indexes, there are now two.
- Matrix:



Recall: `matrix()` Function

- The `matrix()` function may be used to create a matrix.
- Arguments:
 - `nrow = 4` sets the number of rows (four in this case).
 - `byrow = FALSE` fills the matrix by columns.

Matrix Addition and Subtraction

- Matrix addition and subtraction are performed element-by-element.
- Matrices that are added or subtracted need to have equal dimensions.
 - *They do not need to be square.*
- Matrices **are** commutative and associative with respect to addition.
 - Order of terms and location of parentheses does not matter.
- Matrices **are not** commutative and associative with respect to subtraction.
 - Order of terms and location of parentheses **does** matter.

Example 1

- Create the following matrices:
 - `A <- matrix(data = c(3,-1,0,4,5,2), nrow = 3)`
 - `B <- matrix(data = c(4,1,-2,8,-3,6), nrow = 3)`
 - `C <- A + B`
- And do the following calculations:
 - 1 $A + B$
 - 2 $B + A$
 - 3 $(A + B) + C$
 - 4 $A + (B + C)$
 - 5 $A - B$
 - 6 $B - A$
 - 7 $(A - B) - C$
 - 8 $A - (B - C)$
- What did you notice?

Scalar Multiplication

- Element-by-element multiplication is the default in R and Python using `*`.
- Each element of the matrix is simply multiplied by the scalar.
 - This process is commutative, associative, and distributive.
- **As long as the matrices are the same dimensions,** element-by-element multiplication may also be applied.

Example 2

- Create the following matrices and scalars:
 - `A <- matrix(data = c(1,2,3,0,2,0,-5,3), nrow = 2)`
 - `B <- matrix(data = c(1,2,3,0,2,0,-5,3), nrow = 2)`
 - `k <- 3`
 - `c <- 5`
- And do the following calculations:
 - ① `k * A`
 - ② `A * k`

 - ③ `c * (k * A)`
 - ④ `(c * k) * A`

 - ⑤ `k * (A + B)`
 - ⑥ `k * A + k * B`

 - ⑦ `A * A`
 - ⑧ `A * B`
- What did you notice?

Matrix Multiplication I

- Matrix multiplication is more complex and is not possible at times.
 - Used to obtain most analytical solutions we have covered so far.
- **The number of columns in the first matrix must equal the number of rows in the second matrix.**
- Each element in a row of the first matrix is multiplied by the element in the corresponding column and position of the second matrix, and the products are summed together.
- In R we use `%*%` and functions from NumPy in Python.

Matrix Multiplication II

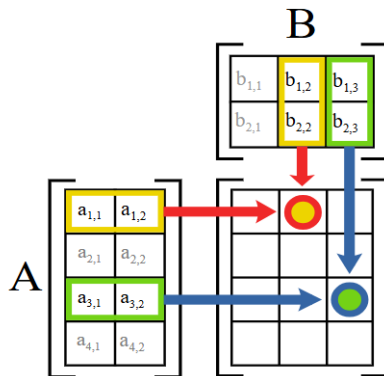


Figure: source: (1)

- Matrix multiplication is associative but not commutative.

Illustrative Example 1

- Multiply the following matrices (**AB**) by hand:

$$\mathbf{A} = \begin{array}{|c|c|} \hline 3 & 1 \\ \hline -2 & 2 \\ \hline 1 & -1 \\ \hline 0 & 2 \\ \hline \end{array} \quad \mathbf{B} = \begin{array}{|c|c|c|} \hline 1 & -2 & 2 \\ \hline 1 & 0 & 3 \\ \hline \end{array}$$

- Note: 4×2 by 2×3 .
- **AB**: pre-multiplication of **B** by **A** or post-multiplication of **A** by **B**.

Example 3

- Using A and B from Illustrative Example 1, do the following calculations:
 - $A \%* \% B$
 - $B \%* \% A$
 - $A \%* \% (B \%* \% B)$
 - $(A \%* \% B) \%* \% B$
- What did you notice?

Matrix Transposition

- **Matrix transposition** is the process of switching the rows of a matrix for the columns.

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -2 & 2 \\ 1 & -1 \\ 0 & 2 \end{bmatrix} \quad \mathbf{A}^T = \begin{bmatrix} 3 & -2 & 1 & 0 \\ 1 & 2 & -1 & 2 \end{bmatrix}$$

- The transpose may be used to facilitate matrix multiplication.
- The $\mathbf{t}(A)$ function is used in R and `numpy.transpose(A)` in Python.

Matrix Symmetry

- A matrix is said to be **square** if the number of rows is equal to the number of columns.
- A square matrix is said to be **symmetric** if it is identical to its transpose.
 - This can be verified by subtracting the transpose of the matrix from the original square matrix ($A - t(A)$).
- *All symmetric matrices are square, but not all square matrices are symmetric.*

Example 4

- Create the following matrices:
 - `A <- matrix(data = c(1,2,3,0,3,-2), nrow = 2)`
 - `B <- matrix(data = c(1,3,-1,-2,2,3,2,2,3), nrow = 3)`
- And do the following calculations:
 - 1 `A %*% B`
 - 2 `B %*% A`
 - 3 `t(A)`
 - 4 `t(B)`
 - 5 `B %*% t(A)`
- What did you notice?

Example 5

① Create the following square matrices:

- `A <- matrix(data = c(1,2,2,2,4,3,2,3,-4), nrow = 3)`
- `B <- matrix(data = c(1,3,-1,-2,2,3,2,2,3), nrow = 3)`

② Calculate the transpose of **A** and **B**.

③ Are either of these symmetrical?

Matrix Inversion

- An **identity matrix** is a symmetric square matrix with a value of 1 on the diagonal and 0 elsewhere.
 - `diag()` function in R and `numpy.identity()` in Python.
- The **inverse** of a matrix **A** (denoted $\mathbf{A}^{-1}$) is a matrix that, when matrix multiplied by **A**, yields an identity matrix of the same dimensions.
 - `solve()` in R and `numpy.linalg.inv()` in Python.
- *Inversions only exist for square matrices, but not all square matrices have an inverse.*
- A matrix is **orthogonal** if its inverse and its transpose are identical.
 - An identity matrix is an example of an orthogonal matrix.

Example 6

- Create the following matrices:
 - `A <- matrix(data = c(1,2,2,2,4,3,2,3,-4), nrow = 3)`
 - `B <- matrix(data = c(1,2,3,2,3,1,3,2,1), nrow = 3)`
- And do the following calculations:
 - 1 The inverse of **A** and **B**
 - 2 `A %*% solve(A)`
 - 3 Is **A** orthogonal?
- What did you notice?

Concluding Thoughts

- These concepts are widely used in much of what we have done and what we will do in this course.
- Distance measures and dimension reductions rely heavily on properties of matrices.
- *Analytical estimates to linear regression models are made through linear algebra.*

Next Lecture

- We will discuss **Distance measures**.
- Eigenanalysis.
- Properties of Distance Measures
- Common Distance Measures

Exercise 1

- Create a 5×5 identity matrix ($\mathbf{I}$) and do the following:
 - ① Calculate the transpose of $\mathbf{I}$.
 - Do you notice anything?
 - ② Calculate $\mathbf{I}^{-1}$.
 - ③ Verify that $\mathbf{I}$ is orthogonal.

References & Resources

- ① Bakker, J. D. (2026) Applied Multivariate Statistics in R
- ② Peng R. D. (2020) *Exploratory Data Analysis with R*
- ③ Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC